External Validation of a Seven-Weight Scorecard versus Tree Ensembles in Ovarian Cancer Risk Prediction

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***Abstract*—Preoperative blood tests help decide whether an ovarian mass requires cancer surgery, and a model that degrades silently at a new hospital can delay surgery or trigger unnecessary operations. We trained eight models (a CA125 cutoff rule, the published formulas ROMA and CPH-I, a logistic regression, a hand-computable integer scorecard, and three tree ensembles) on a Chinese dataset (349 patients) and applied them unchanged elsewhere. The primary external benchmark is 100 West China patients with measured biomarkers (63 cancers); the 74-percent-imputed cohort (380 patients) is co-primary, and a Japanese cohort (150 healthy women, 27 cancers) serves only as a specificity stress test. On the 100 complete cases, the scorecard (area under the ROC curve 0.971) was numerically highest and did not differ significantly from the logistic regression (0.941) or ROMA (0.965). A simulation calibrated to the real data, intended only as an illustrative counterexample, showed that under the simulated concept drift examined here the flexible models' advantage diminished and reversed. Within these small Asian cohorts, the scorecard performed no worse than far more complex ensembles and missed fewer cancers than the off-the-shelf formulas; generalization beyond these cohorts requires larger validation studies.**

***Index Terms*—ovarian cancer; risk prediction; scorecard; external validation; transportability; concept drift**

I. Introduction

Ovarian cancer is the most lethal gynecologic cancer, and whether a pelvic mass goes to a specialist cancer surgeon depends on preoperative risk; getting it wrong in either direction is costly. That risk is assessed from two blood proteins measured cheaply before surgery, CA125 (cancer antigen 125) and HE4 (human epididymis protein 4), combined either by fixed formulas such as the Risk of Ovarian Malignancy Algorithm (ROMA) [1] and the Copenhagen Index (CPH-I) [2] or by a fitted model; proposed machine learning models usually show strong training results but little external testing [3], [4].

This paper asks whether model complexity buys performance that survives a move to a new hospital, and whether a bedside-computable scorecard matches the complex alternatives. We trained eight models on one Chinese cohort, applied them unchanged to a second Chinese cohort and a Japanese specificity stress test, and measured discrimination, calibration, and decision consequences, plus a simulation of when complex models help.

The authors acknowledge the use of Google Gemini, an AI-based assistant, for grammar checks and basic code debugging; the authors retain full authority and responsibility for the design, methodology, analysis, and all claims.

II. Methods

A. Data

Three public, anonymized datasets were used [5]–[7]. Institutional review board approval was not required for reanalysis of public, de-identified data; the cohorts are a training cohort from Changzhou, Jiangsu, China (349 patients; 171 ovarian cancers, 178 benign masses; Mendeley Data, doi:10.17632/th7fztbrv9.11 [8]), a second Chinese cohort from the West China Second University Hospital of Sichuan University, Chengdu (380 patients; 188 cancers, 192 benign cysts; figshare, doi:10.6084/m9.figshare.28831256), and a Japanese cohort from a university hospital in Japan (177 patients; 27 cancers, 150 healthy women; Karger figshare, doi:10.6084/m9.figshare.24235450, [7]). Each patient has four features (CA125 U/mL, HE4 pmol/L, age, menopausal status). The raw files are `chinese training dataset(All Raw Data).csv`, `west_china.xlsx`, and `japanese.xlsx`; `src/clean_data.py` harmonizes them into the three processed files used by every analysis. Scattered missing training biomarker values were filled by feature-wise cohort medians; externally, HE4 was missing for 74% of West China patients and age for 85% of Japan patients, and these gaps were filled with the frozen training-set medians, so no statistic is computed on a validation cohort. The external validation evidence is therefore deliberately narrow. The primary West China analyses use the 100 patients with measured HE4 (63 cancers, 37 benign). Complete-case analysis is primary for three reasons: (i) the alternative, training-median imputation, replaces 74% of HE4 values with one constant, which cannot reflect between-patient variation, forces most patients into a single HE4 bin, and can pull discrimination toward the training distribution, so its narrower uncertainty is artificial; complete-case analysis avoids this bias at the cost of a smaller sample; (ii) multiple imputation by chained equations (MICE, Section III-C) degrades the scorecard's tertile binning (0.853), so imputed benchmarks depend on the imputation method; and (iii) the complete cases evaluate the exact inputs a clinician would measure. The 280 excluded patients (125 cancers, 155 benign) had similar age, CA125, and menopausal rates. The 74%-imputed full cohort ($N = 380$) is reported alongside as a co-primary analysis. Because it contains 150 healthy women and only 27 cancers, the Japanese cohort is treated as a specificity stress test rather than external validation of the triage tool: its AUROC is reported separately in Table II and enters the pooled analysis only as a discrimination sensitivity check.

B. Models

Eight models used the same four features. (1) *Integer scorecard*: each continuous feature was rank-transformed to a uniform score (QuantileTransformer, 100 quantile points) and cut into training-set tertiles, with frozen cutpoints CA125 24.3/165.0 U/mL, HE4 45.8/92.7 pmol/L, and age 37.0/52.0 years; the three bins per feature were one-hot encoded and fitted with an L2-regularized logistic regression ($C = 1.5$). Each coefficient β_i was converted to an integer weight $w_i = \mathrm{round}(5\beta_i/\max_j|\beta_j|) \in \{-5,\ldots,+5\}$, with $|\beta_i| < 0.01$ set to zero; the scaling factor was $5/\max_j|\beta_j| = 2.151$, so the rounding is exactly reproducible. The deployed score sums the integer weights of the patient's bins (range −7 to +9, computable by hand; Table I) and uses seven nonzero weights (HE4 −4/−1/+5, CA125 −1/+2, age −2/+2); the CA125 and age mid-bins round to zero, and the fitted intercept is not used in scoring. The binary menopausal feature, binned identically, collapses to one constant column whose coefficient rounds to zero; it is effectively dropped, not manually removed (its row in Table I documents this). (2) *Logistic regression* on standardized features. (3–4) the published formulas *ROMA* and *CPH-I*, computed on raw values. ROMA uses $\mathrm{PI} = -12.0 + 2.38\ln(\mathrm{HE4}) + 0.0626\ln(\mathrm{CA125})$ premenopause and $\mathrm{PI} = -8.09 + 1.04\ln(\mathrm{HE4}) + 0.732\ln(\mathrm{CA125})$ postmenopause, mapped to a percentage by the logistic transform, with published cutoffs 13.1% (pre) and 27.7% (post) [1]. CPH-I uses $-14.0647 + 1.0649\log_2(\mathrm{HE4}) + 0.6050\log_2(\mathrm{CA125}) + 0.2672\times \mathrm{age}/10$, likewise logistic-transformed, with cutoff 7% [2]. Units: HE4 pmol/L, CA125 U/mL, age years; menopause is the dataset's binary postmenopausal indicator; imputed HE4 uses the frozen training median. Within-cohort recalibration means fitting, on West China, a logistic regression of the outcome on the formula's probability as the sole covariate. (5) a single *CA125 ≥ 35 U/mL* rule. (6–8) *XGBoost*, *CatBoost*, and *Random Forest*, evaluated under two settings: Setting A keeps each package's default hyperparameters, and Setting B tunes each ensemble by internal cross-validation over a 27-configuration grid per family, with all tuning decisions confined to training folds and the external cohorts never entering selection (XGBoost 100 trees, depth 3, learning rate 0.03; CatBoost 500 iterations, depth 4, learning rate 0.03; Random Forest 500 trees, depth 6, minimum leaf size 2). Setting B is primary (Table II; Setting A in the footnote). All models were trained once on the Chinese cohort; every

TABLE I
THE INTEGER SCORECARD, FITTED ON THE CHINESE TRAINING COHORT (349 PATIENTS) AND APPLIED UNCHANGED EXTERNALLY; FEATURES BINNED BY FROZEN TRAINING-SET TERTILES.

Feature	Low bin	Mid bin	High bin
CA125 (U/mL)	−1	0	+2
HE4 (pmol/L)	−4	−1	+5
Age (years)	−2	0	+2
Menopausal status	0	0	0

parameter, cutpoint, and threshold was frozen before external evaluation.

C. Evaluation

Discrimination was measured by the area under the ROC curve (AUROC). Confidence intervals (95%) and two-sided paired tests of between-model differences used 2,000 patient-level bootstrap resamples, which preserve the paired structure across models and keep Monte Carlo error below 0.001 AUROC. The prespecified primary comparison is the scorecard versus ROMA, tested per cohort and summarized by a DerSimonian–Laird random-effects model [9] across the two external cohorts; with only two cohorts the heterogeneity estimate is imprecise, so per-cohort tests are primary and the pooled estimate is a summary. All other comparisons are secondary, and effect sizes with 95% CIs are emphasized over p-values. Calibration was measured with the Brier score and calibration-in-the-large. Decision value used a weighted-harm analysis (Section III-B): thresholds were swept over 1–100%, and for exchange rates $w \in \{1, 5, 10, 50\}$ the threshold minimizing $H = w \times \text{missed} + \text{unnecessary}$ per 100 women was identified. We distinguish Q1 (does each model transport unchanged?) from Q2 (how well can each model perform after local recalibration?). The exchange-rate weights are hypothetical scenarios, not validated clinical utilities, spanning equal valuation of a missed cancer and an unnecessary referral ($w = 1$) to valuing a missed cancer 50 times as much ($w = 50$); they characterize sensitivity to relative error costs. All design choices preceded external evaluation: the four features are those shared by all three cohorts and required by the published formulas (the primary comparator is ROMA, the most-used published formula); the primary comparison (scorecard versus ROMA) and the four weights were prespecified; only the scorecard's bin count and regularization were selected among 12 pre-specified candidates (bins 3/4/5/7 with $C = 0.5/1.5/5.0$) by repeated CV (Section III-E). The primary harm comparison applies within-cohort recalibration to the scorecard and the two formulas (the logistic regression and XGBoost are shown as fitted); published-cutpoint operating points are secondary, and no pooled decision analysis was performed. The retraining experiment used stratified training subsets ($n = 60$ to 349, 10 draws each), evaluated unchanged on the imputed West China cohort. Robustness was tested with 10% and 30% multiplicative, additive, and combined noise, 5% gross-error contamination, and ±10% bin-boundary shifts.

D. Implementation and availability

Analyses used Python 3.13 with scikit-learn 1.7.2, XGBoost 3.3.0, CatBoost 1.2.10, and Random Forest, all with random seed 42; cross-validation used stratified five-fold splits. Setting A kept the package defaults (XGBoost 100 trees, depth 6, learning rate 0.3; CatBoost 1,000 iterations, depth 6, learning rate 0.03; Random Forest 100 trees, no depth cap); Setting B used the internal-CV-selected configurations of Section II-B. The complete code and results are available at https://github.com/angelayaoo/Ovarian-Cancer-RiskPrediction, with a one-command pipeline (`reproduce.sh`) and pinned dependencies.

E. Simulation

The simulation is an illustrative counterexample, not an explanation of the real-world results, and no general claim about real concept drift is made from it. A simulation calibrated to the Chinese data drew biomarkers from class-conditional lognormal distributions and generated the cancer label from the logistic model

$$\begin{aligned}\eta(z_1, z_2, z_3) = {} & -0.65 + 1.8z_1 + 1.0z_2 + 0.5z_3 \\ & + (1-d)\big(1.6z_1z_2 + 1.2\sin(1.7z_1)\big),\end{aligned} \tag{1}$$

where z_1, z_2, and z_3 are standardized log CA125, log HE4, and age, and $d \in [0,1]$ sets the drift: $d = 0$ keeps the interaction and nonlinear terms, $d = 1$ removes them. Clinically, $d = 1$ approximates a hospital where how the markers combine with age differs—for example, different assay calibration across laboratories—so interactions learned in one cohort no longer hold; it is a stylized proxy, not a model of any hospital. The grid crossed training sizes $n \in \{100, 200, 400, 800\}$, shifts $s \in \{0, 0.5, 1.0\}$, noise $\{0\%, 50\%\}$, and drift $d \in \{0, 1\}$, with 20 replicates per cell.

III. Results

A. Discrimination

On the 100 West China patients with measured HE4 (63 cancers; the primary external benchmark), the scorecard was numerically highest, with ROMA, tuned XGBoost, and the logistic regression within overlapping CIs (Table II, West CC column; Fig. 1).

On the co-primary 74%-imputed cohort ($N = 380$) the ranking was similar (logistic regression 0.936, scorecard 0.929, ensembles 0.877–0.895).

As a discrimination benchmark, a random-effects (DerSimonian–Laird) meta-analysis with paired bootstrap tests pooling the imputed West China cohort and the Japanese stress-test cohort gave a pooled scorecard-minus-ROMA difference of +0.043 AUROC (95% CI 0.013 to 0.074; $p = 0.006$). The scorecard also exceeded each tuned tree ensemble (pooled differences +0.036, +0.055, and +0.033 for XGBoost, CatBoost, and Random Forest; all $p \leq 0.016$) and did not differ from the logistic regression (−0.005; $p = 0.55$). After Holm–Bonferroni correction across the five prespecified pooled comparisons, the scorecard–ROMA difference remained significant (adjusted $p = 0.023$), as did

TABLE II
AUROC (95% CI) by model and cohort.

Model	West CC[a]	West[b]	Japan[c]
Scorecard	**0.971** 0.936–0.995	**0.929** 0.902–0.955	0.845 0.714–0.949
Logistic reg.	0.941 0.887–0.983	**0.936** 0.908–0.961	0.806 0.666–0.926
ROMA	0.965 0.930–0.992	0.886 0.848–0.921	0.799 0.666–0.908
CPH-I	0.929 0.875–0.973	0.868 0.828–0.905	**0.920** 0.852–0.973
CA125 rule	0.694 0.605–0.785	0.642 0.597–0.684	0.781 0.685–0.874
XGBoost	0.946 0.897–0.984	0.890 0.853–0.925	0.840 0.711–0.944
CatBoost	0.910 0.845–0.962	0.877 0.836–0.915	0.772 0.650–0.880
Random Forest	0.930 0.871–0.979	0.895 0.858–0.931	0.819 0.692–0.926

[a]n=100, measured-HE4 complete cases (primary); [b]N=380, 74%-imputed (co-primary); [c]n=177, 150 healthy women and 27 cancers (specificity stress test). CC: complete case; CI: confidence interval. Tree ensembles are Setting B (CV-tuned); Setting A (defaults) scored 0.921/0.922/0.910 (CC), 0.887/0.879/0.879 (West), 0.736/0.750/0.794 (Japan).

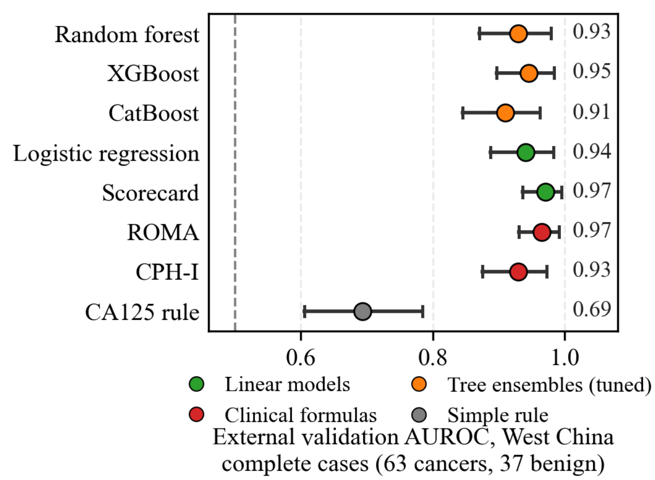

Fig. 1. External validation AUROC, 100 West China patients with measured HE4 (63 cancers); dots show bootstrap means and 95% CIs (2,000 resamples).

all three ensemble comparisons (adjusted p at most 0.031). These pooled comparisons combine the imputed cohort with a stress-test cohort and are secondary to the complete-case benchmark above.

In the Japan specificity stress test ($n = 177$, of whom only 27 have cancer), CPH-I was highest (0.920; Table II, Japan column). It is strictly a specificity stress test, not external validation: healthy volunteers, not women with masses, so its AUROC is not a standard external-validation metric.

B. Decision consequences

In the primary weighted-harm analysis (Q2; Table III, Fig. 2), the scorecard and two formulas were within-cohort recalibrated (logistic regression and XGBoost as fitted). The

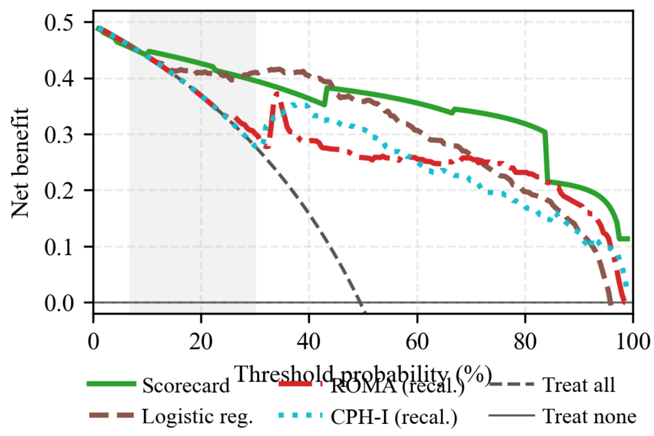

Fig. 2. Decision curve analysis on West China (N = 380): net benefit against the decision threshold, with treat-all (dashed) and treat-none (horizontal axis) references; the shaded band marks the range of published cutpoints (7–30%). Scorecard and formulas are shown after within-cohort recalibration (logistic regression as fitted); XGBoost omitted.

TABLE III
MINIMUM NET HARM ON WEST CHINA (N = 380) AT EXCHANGE-RATE WEIGHT w: HARM = $w\times$MISSED+UNNECESSARY PER 100 WOMEN, MINIMIZED OVER SWEPT THRESHOLDS. Q2 ROWS: SCORECARD, ROMA, CPH-I WITHIN-COHORT RECALIBRATED (LOGISTIC REGRESSION AND XGBOOST AS FITTED). Q1 PUBLISHED-SCALE HARMS: ROMA 15.0/43.4/45.8/45.8; CPH-I 15.0/51.3/64.7/159.5.

Model	w=1	w=5	w=10	w=50
Logistic reg.	**9.7**	**29.2**	**39.7**	46.6
Scorecard	12.1	30.5	45.0	59.5
ROMA (recalibrated)	14.7	42.9	45.8	**45.8**
CPH-I (recalibrated)	15.0	50.3	50.3	50.3
XGBoost	14.7	44.7	66.1	215.5
Treat all (reference)	50.5	50.5	50.5	50.5

scorecard's minimum harm was below recalibrated ROMA and CPH-I at weights up to 5 (12.1 vs. 14.7 and 15.0 at w = 1); at w = 50 both fell below it (45.8 and 50.3 vs. 59.5). The logistic regression was lowest at weights up to 10; at w = 50, recalibrated ROMA was lowest (45.8). Reported minima are over the swept thresholds; where a minimum exceeds treat-all (e.g., XGBoost at w = 50), treating everyone dominates (Table III).

As the Q1 decision consequence, the scorecard's high-risk tier (score $\geq$ +1) missed 47 of 188 cancers with 5 false positives, versus 82 (ROMA) and 48 (CPH-I) at published cutpoints.

C. Calibration and missing-data checks

For calibration on the imputed West China cohort (N = 380), the scorecard's logistic mapping was refit on that same cohort and evaluated there, so its calibration figures are apparent, not prospective; the logistic regression is shown frozen from training and the formulas at published scales, so only the scorecard's figures can be inflated by the shared data. The recalibrated scorecard had Brier 0.097, calibration slope 1.00, and expected calibration error (ECE) 0.042; the logistic regression had Brier 0.111, slope 1.44, ECE 0.135; ROMA had Brier 0.209, slope 1.53, ECE 0.220; CPH-I had Brier 0.259, slope 1.11, ECE 0.303, with both formulas over-predicting risk (calibration-in-the-large −0.95 and −1.42).

Multiple imputation by chained equations (MICE) preserved the flexible-model ranking (logistic regression 0.926) but degraded the scorecard's binning (0.853; see Section II-A).

D. Robustness and simulation

Under 30% multiplicative noise on CA125 and HE4 (West China, N = 380; 100 perturbations), mean AUROC losses were 0.039 (scorecard), 0.002 (logistic regression), 0.073 (ROMA), and 0.084 (XGBoost).

In the simulation (Fig. 3), at 50% noise the ensemble-average minus-scorecard gap averaged +0.018 at d = 0 and −0.018 at d = 1 of (1); at d = 1 the scorecard had the highest AUROC in 22 of 24 cells. Across factors, the gap spread most across drift levels (0.033), then distribution shift (0.012), training size (0.003), and noise (0.002).

E. Internal validity and sensitivity

Across 20 repeated five-fold cross-validations, the scorecard's internal AUROC was 0.906 (SD 0.003) and the three-bin configuration was selected in 18 of 20 repeats; in-sample ensembles fitted 0.094–0.117 above their CV AUROC.

IV. DISCUSSION AND CONCLUSION

Three results follow, within the limits of the available cohorts. First, on discrimination: seven integer weights performed no worse than models with thousands of parameters; the scorecard was numerically best on the 100 complete cases (0.971), similar to the logistic regression on the imputed cohort, and CPH-I won the Japan stress test (0.920). Second, on decisions: the published cutpoints missed 82 of 188 cancers versus 47 for the locally fitted scorecard, whose minimum harm stayed below the recalibrated formulas up to 5:1. Third, under the simulated form of concept drift examined here, the flexible models' advantage diminished and reversed; the simulation alone does not show this reversal occurs in real hospitals.

The primary constraint on generalizability is sample size. (i) The external evidence rests on 100 complete cases (63 cancers) in West China and 27 cancers in the Japanese stress test, and all three cohorts are Asian hospital series; these analyses reduce but do not remove this uncertainty, so the conclusions may not generalize. (ii) HE4 was missing for 74% of West China patients and age for 85% of Japan patients, so much of the external comparison rests on training-set constants; and the Japanese cohort consists of healthy volunteers rather than women with masses, so its discrimination is not directly comparable with West China's; both narrow how broadly the findings apply. (iii) Ultrasound-based models could not be tested (the public data lack imaging), and the simulation is a simplification, not a proof about any specific hospital.

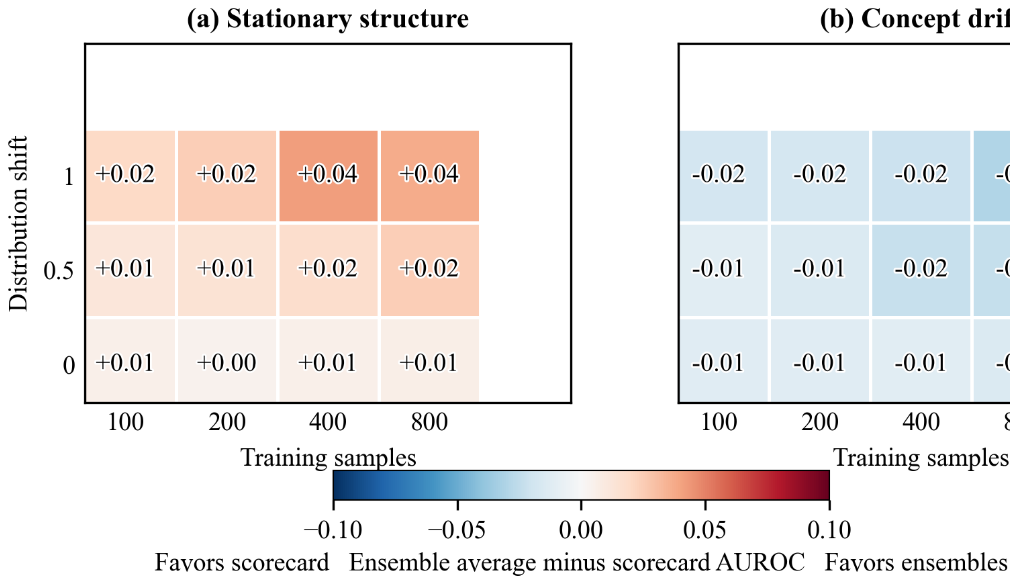

Fig. 3. Regime map from the simulation: mean over 20 replicates of the tree-ensemble-average minus scorecard AUROC at 50% noise (red favors ensembles, blue scorecard); axes: training samples (100–800), distribution shift (0–1).

Future work should add ultrasound variables and validate prospectively in larger, multi-center cohorts. Imaging features (e.g., solid components or ascites) could be binned and weighted exactly like the biomarkers, keeping the score hand-computable with only the new weights refit; on these small cohorts the scorecard already performed no worse than far more complex models.

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